Exercises

Exercise 6.1 · Euclidean divisions.

1. Carry out the Euclidean division of 2,026 by 17.
2. Carry out the Euclidean division of -47 by 5, then of -35 by 4.
3. Determine the natural numbers n whose quotient in the division by 6 equals the remainder.

Solution.

1. $2{,}026 = 17 \times 119 + 3$: quotient 119, remainder 3.
2. $-47 = 5 \times (-10) + 3$ and $-35 = 4 \times (-9) + 1$.
3. $n = 6q + q = 7q$ with $0 \leq q < 6$: $n \in \{0, 7, 14, 21, 28, 35\}$.

Exercise 6.2 · Remainders of powers.

1. Determine the remainder of 3^{100} in the division by 5.
2. Determine the remainder of $5^{2026} + 2^{2026}$ in the division by 3.

Solution.

1. $3^4 = 81 \equiv 1 \pmod 5$, so $3^{100} = \left(3^4\right)^{25} \equiv 1 \pmod 5$: the remainder is 1.
2. $5 \equiv 2 \equiv -1 \pmod 3$, so $5^{2026} + 2^{2026} \equiv 2 \times (-1)^{2026} = 2 \pmod 3$: the remainder is 2.

Exercise 6.3 · Sums of two squares.

1. Prove that for every integer x, $x^2 \equiv 0$ or $x^2 \equiv 1 \pmod 4$.
2. Deduce that an integer congruent to 3 modulo 4 is not the sum of two squares of integers.
3. Can the integer 2,027 be written $x^2 + y^2$ with x, y integers?

Solution.

1. Table: $x \equiv 0, 1, 2, 3 \pmod 4$ gives $x^2 \equiv 0, 1, 4, 9$, that is $0, 1, 0, 1 \pmod 4$.
2. $x^2 + y^2$ is congruent to $0 + 0$, $0 + 1$ or $1 + 1$, hence to 0, 1 or 2 modulo 4, never to 3.
3. $2{,}027 = 4 \times 506 + 3$: no.

Exercise 6.4 · Divisibility test for 11.

1. Prove that $10 \equiv -1 \pmod{11}$, then that $10^k \equiv (-1)^k \pmod{11}$.
2. Deduce that an integer is divisible by 11 if and only if the alternating sum of its digits, starting from the units, $a_0 - a_1 + a_2 - \dots$, is.

3. Is the integer 918,082 divisible by 11?

Solution.

1. $10-(-1)=11$. Compatibility with powers gives $10^k \equiv (-1)^k$.
2. $N=\sum a_k 10^k \equiv \sum a_k(-1)^k \pmod{11}$; N and the alternating sum have the same remainder.
3. $2-8+0-8+1-9=-22$, a multiple of 11: yes ($918{,}082 = 11 \times 83{,}462$).

Exercise 6.5 · Divisibility for every integer. Prove that for every natural number n:

1. 4^n-1 is divisible by 3;
2. n^3-n is divisible by 6.

Solution.

1. $4 \equiv 1 \pmod 3$, so $4^n \equiv 1$ and $3 \mid 4^n-1$.
2. Table modulo 6: for $n \equiv 0,1,2,3,4,5$, $n^3-n \equiv 0,0,6,24,60,120$, all multiples of 6. So $n^3-n \equiv 0 \pmod 6$ for every n. Remark: $n^3-n=(n-1)n(n+1)$ is a product of three consecutive integers; chapter 7 will show why being divisible by 2 and by 3 is enough.

Exercise 6.6 · Congruence equations.

1. Solve $3x \equiv 5 \pmod 7$. First look for an integer u such that $3u \equiv 1 \pmod 7$.
2. Solve $x^2 \equiv 2 \pmod 7$ with a table.

Solution.

1. $3 \times 5 = 15 \equiv 1 \pmod 7$. Multiplying by 5: $3x \equiv 5$ implies $x \equiv 25 \equiv 4 \pmod 7$; conversely $3 \times 4 = 12 \equiv 5$. The solutions are the integers $x = 4+7k$, $k \in \mathbb{Z}$.
2. For $x \equiv 0,1,\dots,6 \pmod 7$: $x^2 \equiv 0,1,4,2,2,4,1$. The solutions are the integers congruent to 3 or 4 modulo 7.

Exercise 6.7 · The last two digits. Determine the last two digits of the decimal expansion of 7^{2026}.

Solution. We work modulo 100. $7^2 = 49$ and $7^4 = 2{,}401 \equiv 1 \pmod{100}$. Since $2{,}026 = 4 \times 506 + 2$, $7^{2026} \equiv 7^2 = 49 \pmod{100}$: the expansion ends in 49.

Problem

Problem · The check digit of an ISBN-13 code. An ISBN-13 code is a sequence of thirteen digits $a_1 a_2 \ldots a_{13}$. Let

$$S = a_1 + 3a_2 + a_3 + 3a_4 + \cdots + 3a_{12} + a_{13},$$

with the weights 1 and 3 alternating. The code is valid if and only if $S \equiv 0 \pmod{10}$; the last digit a_{13}, called the check digit, is chosen to make it so.

1. Consider the fictitious number 978-2-0601-2345-a_{13}. Compute the check digit a_{13}.
2. Assume that a single digit a is replaced by a digit $a' \neq a$.
   (a) Prove that S changes by $w(a' - a)$, with $w \in \{1, 3\}$.
   (b) Check that $7 \times 3 \equiv 1 \pmod{10}$ and deduce that $3(a' - a) \equiv 0 \pmod{10}$ implies $a' - a \equiv 0 \pmod{10}$.
   (c) Conclude: an error in a single digit is always detected.
3. Assume that two distinct adjacent digits a and b are swapped.
   (a) Prove that S changes by $2(b - a)$ or by $2(a - b)$.
   (b) Determine under which condition on a and b the error is not detected. Give an example.

Solution · Problem.

1. The first twelve digits 9, 7, 8, 2, 0, 6, 0, 1, 2, 3, 4, 5 give $9+21+8+6+0+18+0+3+2+9+4+15 = 95$. We need $95 + a_{13} \equiv 0 \pmod{10}$ with $0 \leq a_{13} \leq 9$: $a_{13} = 5$.
2. (a) The term wa becomes wa', the other terms do not change.
   (b) $21 \equiv 1 \pmod{10}$. If $3(a' - a) \equiv 0$, multiplying by 7: $a' - a \equiv 21(a' - a) \equiv 0 \pmod{10}$. The case $w = 1$ is immediate.
   (c) $a' - a$ is a multiple of 10 between -9 and 9, hence zero: a contradiction with $a' \neq a$. The change $w(a' - a)$ in the sum is not a multiple of 10; since the original code was valid (sum a multiple of 10), the new sum is no longer a multiple of 10, and the error is detected.
3. (a) The two digits have weights 1 and 3. The contribution $a + 3b$ becomes $b + 3a$ (or the reverse), a change of $2(a - b)$ or of $2(b - a)$.
   (b) The error is not detected if and only if $10 \mid 2(a - b)$, that is $5 \mid a - b$: $|a - b| = 5$. Example: swapping the adjacent digits 1 and 6 does not change S modulo 10.

Teacher note · Status of the problem. Check codes (ISBN, bank account keys, barcodes) are a "possible problem" of the syllabus. Question 2b uses an inverse modulo 10: the notion is formalised in chapter 7. The number in question 1 is given as an example, for the computation.